Case Document: Magnus's AFO Design & Control Innovation Challenge

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PATIENT CASE: Magnus, 45 years old

Background

- **Condition:** Right hemisphere stroke, 3 years ago. Completed rehabilitation
- **Occupation:** Part-time librarian: requires walking, reaching, navigating stairs
- **Living situation:** House with 14 stairs to entrance (no elevator)
- **Transportation:** Drives adapted vehicle
- **Goals:**
    - Walk faster to keep up with wife and teenage kids
    - Reduce daily fatigue
    - Return to weekend hiking on easy trails

Clinical Assessment - Right Side (Affected)

Muscle Strength (Manual Muscle Testing)

Ankle:

- Dorsiflexors: **Grade 2** (trace activity during swing but cannot resist gravity)
- Plantarflexors: **Grade 3** (can plantarflex foot against gravity but not resist any added resistance)
- Moderate spasticity in plantarflexors

Knee:

- Extensors: **Grade 4** (functional but weaker than left)
- Flexors: **Grade 4**

Hip:

- Flexors: **Grade 4**
- Extensors: **Grade 4**
- Abductors: **Grade 4** (minimal Trendelenburg gait)

Current Gait Pattern (with posterior leaf spring AFO)

Observed Deviations:

- **Swing phase:** Foot drop → compensatory circumduction and hip hiking to clear floor
- **Initial contact:** On forefoot (no eccentric dorsiflexor control)
- **Stance phase:** Mild knee hyperextension in midstance
- **Push-off:** Reduced/absent power generation all joints, particularly ankle

Functional Limitations:

- Walking speed: 0.8 m/s (normal adult: ~1.3 m/s)
- Walking distance: 800m before significant fatigue
- Stairs: Requires handrail, uses step-to pattern (not reciprocal)
- Energy cost: High (visible effort, breathing heavily after 10 minutes)

Current Orthosis

- **Type:** Posterior leaf spring AFO (worn for 2 years)
- **Function:** Prevents some foot drop during swing
- **Limitations:** Does not provide push-off assistance or induce initial contact at heel
- **Patient feedback:** "Comfortable but I'm ready for something better"

Requirements & Constraints

Budget:

- 50,000 SEK (Swedish healthcare subsidy)

Functional Requirements:

- Must work on stairs (both ascending and descending)
- Must fit with normal shoes/boots
- Must be safe if device fails

Patient Preferences:

- Open to technology
- Concerned about appearance ("not too robotic" at work)
- Willing to spend 15-20 minutes donning/doffing
- Interested in device that could improve over time

GROUP ACTIVITY

PHASE 1: Passive AFO Optimization (10 minutes)

Your Task:

Before considering active devices, optimize Magnus's passive AFO.

Questions to Answer:

1. **What are the specific limitations of his current posterior leaf spring AFO?**
    - What gait problems does it NOT address?
2. **Which passive AFO design would you recommend?**
    - Hinged with plantarflexion stop?
    - Solid/near-solid AFO?
    - Carbon fiber spring ankle joint?
    - Other?
3. **What specific gait improvements would you expect?**
    - What kinematic changes?
    - What kinetic changes?
    - What temporospatial changes?
4. **What are the trade-offs of your passive solution?**
    - Cost vs. benefit
    - Function vs. appearance
    - What problems remain unsolved?

Prepare to share: Your AFO choice and primary rationale (max 30 seconds)

PHASE 2: Active Ankle Exoskeleton Control Design (20 minutes)

The Scenario:

Magnus has tried your optimized passive AFO (assume it's the carbon fiber spring design). It's better—improved push-off and energy return. **BUT significant problems remain:**

- Still initial contact at toe/forefoot (no active dorsiflexion control)
- Cannot adapt to different terrains (stairs, slopes, uneven ground)
- Energy return is fixed (cannot adjust for fatigue or activity level)
- Cannot provide more assistance when needed (end of day, stairs)

Your research team has received funding to develop an active ankle exoskeleton specifically for stroke survivors like Magnus.

Core Control Problem (ALL GROUPS)

Design a control system that detects Magnus's movement intentions and responds appropriately for:

✓ **Level walking** - dorsiflexion at initial contact, stance stability, push-off assistance
✓ **Ascending stairs** - greater push-off power, different timing than level walking
✓ **Descending stairs** - controlled plantarflexion (safety critical!)
✓ **Sit-to-stand transitions** - system must not activate inappropriately
✓ **Activity transitions** - how does the system detect Magnus is approaching stairs?

Critical Constraints:

- **Response time:** Must be <150ms (longer delays cause instability and trips)
- **Safety:** Must fail to passive/safe mode if system errors or battery dies
- **Adaptation:** Should improve/personalize over time as Magnus adapts to the device

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GROUP-SPECIFIC CHALLENGES

GROUP A: Multi-Modal Sensor Fusion

Your Focus: Combining multiple sensor types for robust control

Available Sensors:

- EMG (surface electrodes on muscles)
- IMUs - Inertial Measurement Units (accelerometers, gyroscopes)
- Pressure sensors (in shoe insole)
- Joint angle sensors (potentiometers, encoders)
- Force sensors (ground reaction force)
- Optional: Camera/vision system (smartphone-based)

Your Challenge:

1. Which sensors will you use? Where will you place them?
2. How will you fuse/combine the data from multiple sensors?
3. How will you handle conflicting information between sensors?

Consider:

- EMG alone is noisy and affected by spasticity (false signals)
- IMUs can detect movement that's already happening (not intent)
- How do you detect stairs BEFORE Magnus reaches them?
- Could you use left leg sensors to predict right leg needs?
- Pressure patterns change before activity transitions—how to use this?

GROUP B: Intent Detection & Prediction

Your Focus: Detecting what Magnus wants to do before he does it

The Challenge of Prediction:

- What happens in the body 200-500ms BEFORE heel strike?
- How do you detect the difference between "walking faster" and "about to climb stairs"?
- How do you know if Magnus wants to stop walking vs. just slowing down?

Your Challenge:

1. What signals indicate Magnus's intent before movement begins?
2. How can you use his unaffected (left) leg as a predictor?
    - Example: Left heel strike → right leg toe-off coming in ~300ms
3. How would you detect different activities (level, stairs up, stairs down, stopping)?
4. Could the system learn his personal movement patterns over weeks/months?

Consider:

- Early signals: weight shift, preparatory EMG, contralateral leg timing
- Context clues: time of day (morning = fresh, evening = fatigued), location patterns
- Machine learning: system adapts to Magnus's unique patterns

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GROUP C: Adaptive Assistance Levels

Your Focus: Determining how much the exoskeleton should help

The Assistance Dilemma:

- Too much help → Magnus's muscles weaken further (learned non-use)
- Too little help → Device provides no benefit
- Fixed assistance → Can't adapt to changing needs

Your Challenge:

1. How much assistance should the device provide? (0-100%?)
2. How should assistance level change based on:
    - Time of day (morning vs. evening fatigue)
    - Terrain (flat ground vs. stairs)
    - Walking speed (slow vs. fast)
    - Magnus's effort level
3. How do you prevent muscle weakening from over-assistance?
4. Could human-in-the-loop optimization personalize assistance? (Lecture Slide 25)

Consider:

- "Assist-as-needed" strategy: provide minimum assistance required
- How do you measure when Magnus is struggling vs. not trying?
- Should the device encourage him to use residual muscle strength?
- Rehabilitation goals vs. compensation goals
- Could the system gradually reduce assistance as Magnus improves?

GROUP D: Safety & Failure Modes

Your Focus: What happens when things go wrong

Potential Failures:

- Misclassifying activity (thinking stairs-down is level walking → fall risk)
- Actuator activates during swing phase → trip risk
- Battery depletes mid-stairs → loss of support
- Spasticity triggers false EMG signal → inappropriate activation
- Sensor malfunction or detachment
- User does something unexpected

Your Challenge:

1. Identify the 3 most dangerous failure modes for Magnus
2. How does your system detect when it's "confused" or uncertain?
3. What is your fail-safe strategy for each failure mode?
4. Should there be manual override controls? Emergency stop button?

Consider:

- Some failures are annoying (device stops working)
- Some failures are dangerous (device does wrong thing)
- How do you prioritize safety without making device overly conservative?
- "Do no harm" principle: device should never make Magnus less safe than without it
- How to balance autonomy vs. safety?
- Alert systems: warnings to user when system is uncertain?

DELIVERABLES

Create a sketch/diagram showing:

1. **System overview:**
    - What sensors are you using and where are they placed?
    - Main hardware components
2. **Control logic:**

    - Simple flowchart showing decision-making process
    - How do inputs lead to actions?
3. **How your system addresses:**
    - Level walking
    - Ascending stairs
    - Descending stairs
    - Transitions between activities
    - **YOUR GROUP'S SPECIFIC FOCUS AREA**
4. **Key innovation:**
    - What is novel/clever about your approach?
    - Why is it better than existing solutions?

Be prepared to answer:

- What could go wrong with your approach?
- What's your backup plan when it does?
- Could this realistically be developed in 5-10 years?
- What existing technology can you build on?

PRESENTATION (1.5 minutes per group)

What to cover:

1. Your control approach (30 seconds)
2. Key innovation for your focus area (30 seconds)
3. Main limitation/challenge you identified (30 seconds)

Class Discussion:

- Could approaches from different groups be combined?
- Which innovation would have the biggest impact on Magnus's daily function?
- What's the gap between your ideas and current technology?
- Could human-in-the-loop optimization help?

Remember: Magnus could easily represent a real person with real goals. Your solution should improve his life in meaningful ways.
